Telecom Signal Stream

Data Forge

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Problem Statement

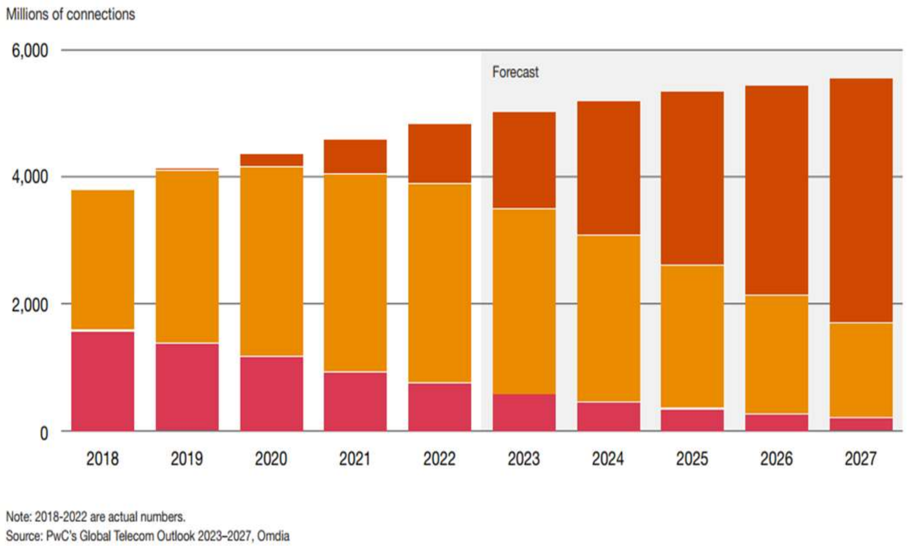
- Telecom companies generate huge volumes of mobile-signal data every day.
- Delayed or manual analysis slows down issue detection and response.
- Need a real-time solution to monitor and analyse network health instantly.

Objective

- Build a real-time analytics pipeline using AWS Kinesis, Glue, Athena, and CloudWatch.
- Enable continuous tracking of key metrics like signal strength, GPS precision, and network status.

Use Case

- Simulate mobile log data (CSV) as live stream via Kinesis.
- Perform ETL and cleaning using AWS Glue.
- Store curated data in S3 and query insights with Athena.
- Key Metrics:
  - Avg. Signal Strength per Region
  - GPS Accuracy per Postal Area and Operator
  - Network activity (Still, Tilting, Unknown, On_Foot, In_Vehicle)

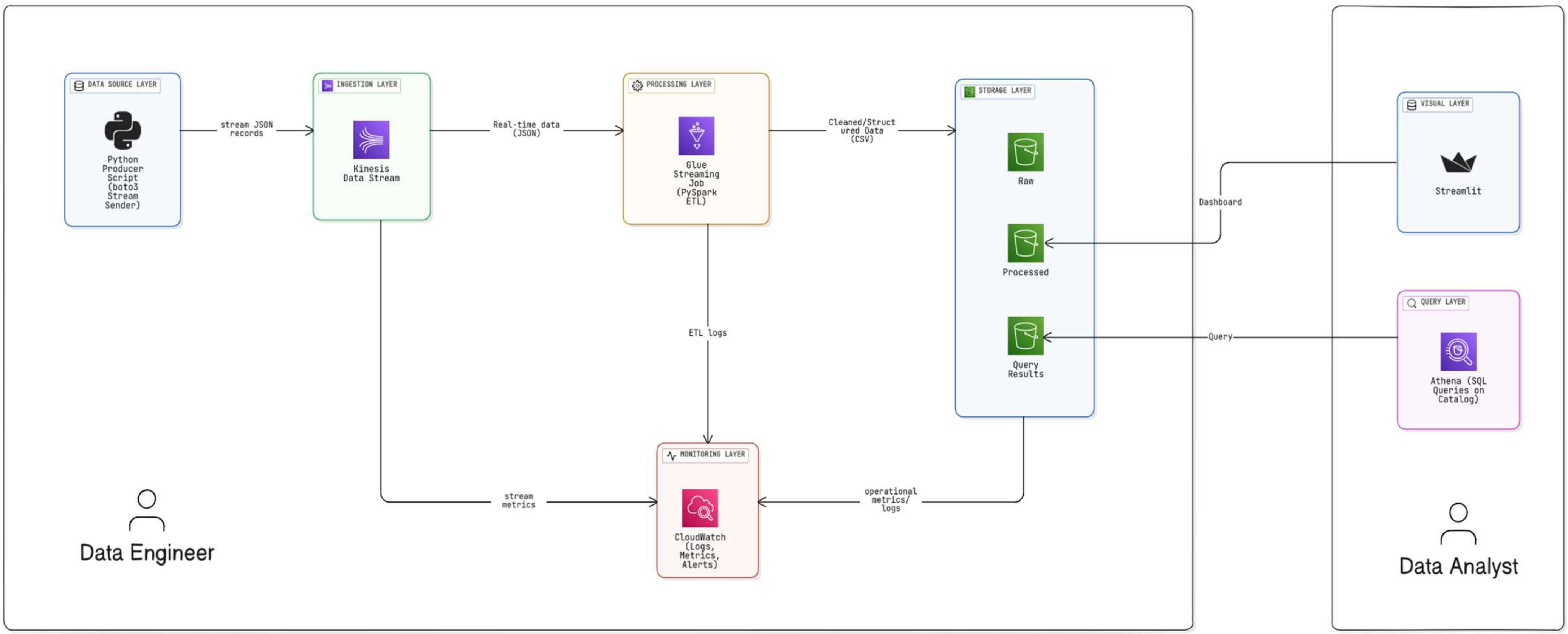


Input Data Overview

- Schema & Sample
  - Dataset : mobile_logs
  - Format – csv
  - Streaming Data

hour	lat	long	signal	network	operator	status	description	speed	satellites	precision	provider	activity
00:36:07.000	9.7028	-16.9857	15	DTAC	JAZZTEL	0	STATE_IN_SERVICE	35.6	5.6	144.1	fused	STILL
00:36:08.000	3.68379	-2.76099	22	movistar	DTAC	0	STATE_IN_SERVICE	128.9	6.3	14.6	fused	STILL
00:36:09.000	28.02161	-3.30627	16	orange	RACC	0	STATE_IN_SERVICE	134.9	2.6	126.4	fused	UNKNOWN
00:36:10.000	16.51063	-9.35435	6	orange	vodafone ES	0	STATE_IN_SERVICE	45.3	11	139.9	fused	UNKNOWN
00:36:11.000	16.803	-12.9745	6	vodafone	Orange	0	STATE_IN_SERVICE	123	14.7	28	gps	IN_VEHICLE

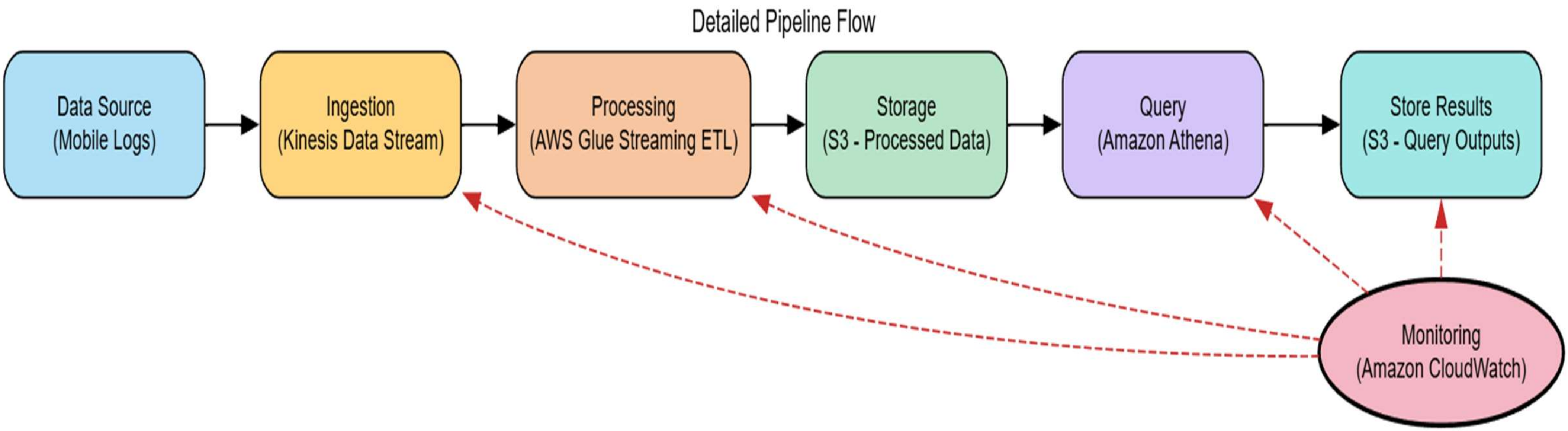
High-Level Architecture



Tools & Services Used:

Service	Purpose
Amazon S3	Central data lake (stores raw, processed, scripts, checkpoints, and query outputs).
Amazon Kinesis Data Stream	Real-time ingestion of mobile log data (acts as the streaming data source).
AWS Glue	Performs ETL processing (Spark Streaming job reads from Kinesis, writes to S3 in Parquet).
AWS IAM	Provides roles and permissions for Glue, Kinesis, and S3 access control.
Amazon Athena	Serverless query engine to analyze processed data stored in S3 using SQL.
Amazon CloudWatch	Monitors Glue job execution and logs for troubleshooting.

Detailed Pipeline Flow



Data Ingestion

- **Python Data Ingestion Script**

- Simulates real-time telecom data from CSV.
- Converts each row to JSON and sends it to Kinesis Data Stream.

```python
import csv, json, time, boto3

STREAM_NAME = "kinesis-mobile-log-stream"
CSV_PATH = "mobile-logs.csv"  # user's dataset path
SLEEP = 0.05

kinesis = boto3.client("kinesis", region_name="us-east-1")

with open(CSV_PATH, 'r') as f:
    reader = csv.DictReader(f)
    for i, row in enumerate(reader):
        data = json.dumps(row)
        kinesis.put_record(StreamName=STREAM_NAME, Data=data.encode('utf-8'), PartitionKey=str(i%10))
        if i % 50 == 0:
            print("sent", i)
        time.sleep(SLEEP)
```

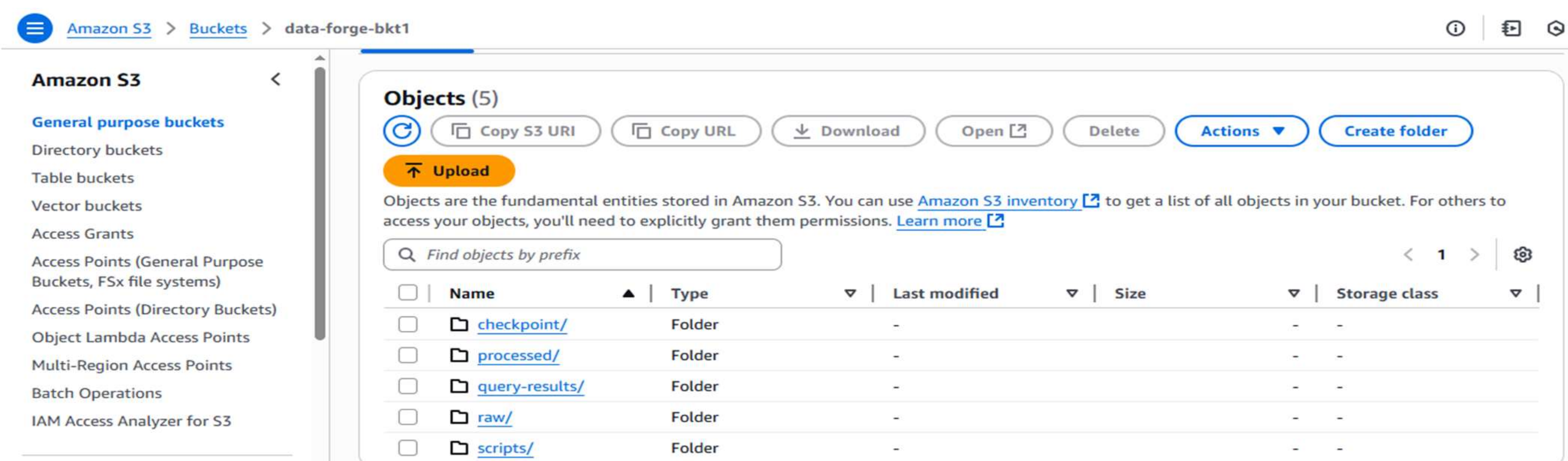
```
sent 0
sent 50
sent 100
sent 150
sent 200
sent 250
sent 300
sent 350
```

S3 Bucket Setup

```python
try:
    # 1. Create the S3 bucket
    s3_client.create_bucket(Bucket=bucket_name)
    print(f"Bucket '{bucket_name}' created.")
except ClientError as e:
    print(f"Error creating bucket: {e}")
```

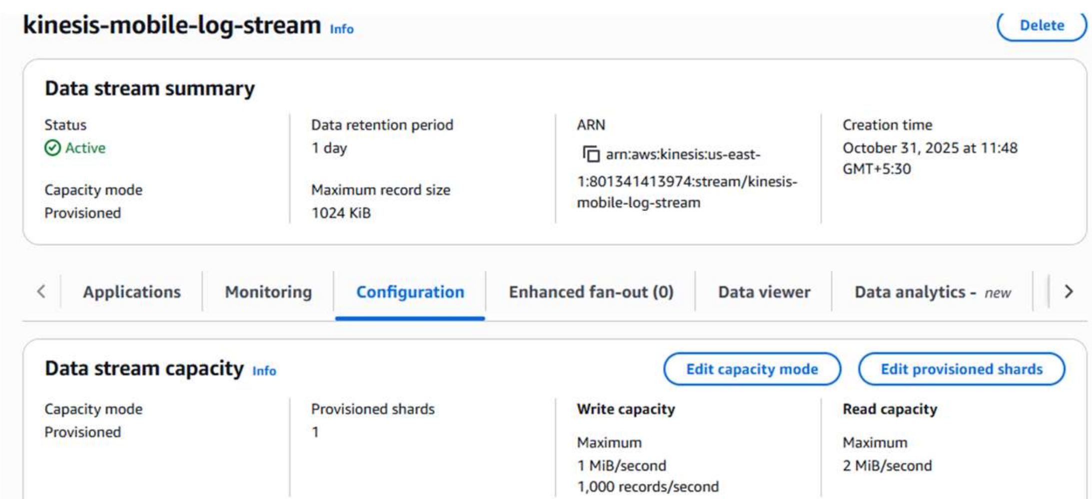
```python
folders = ['raw/', 'processed/', 'scripts/', 'checkpoint/', 'query-results/']

for folder in folders:
    try:
        s3_client.put_object(Bucket=bucket_name, Key=folder)
        print(f"Folder '{folder}' created.")
    except ClientError as e:
        print(f"Error creating folder '{folder}': {e}")
```



Kinesis Data Stream

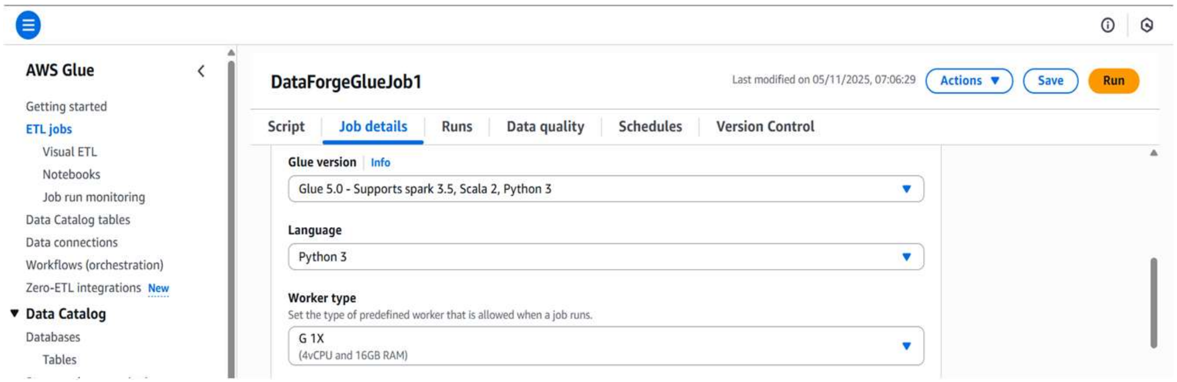
Serverless Amazon Web Services (AWS) service for processing and analyzing real-time streaming data at scale.



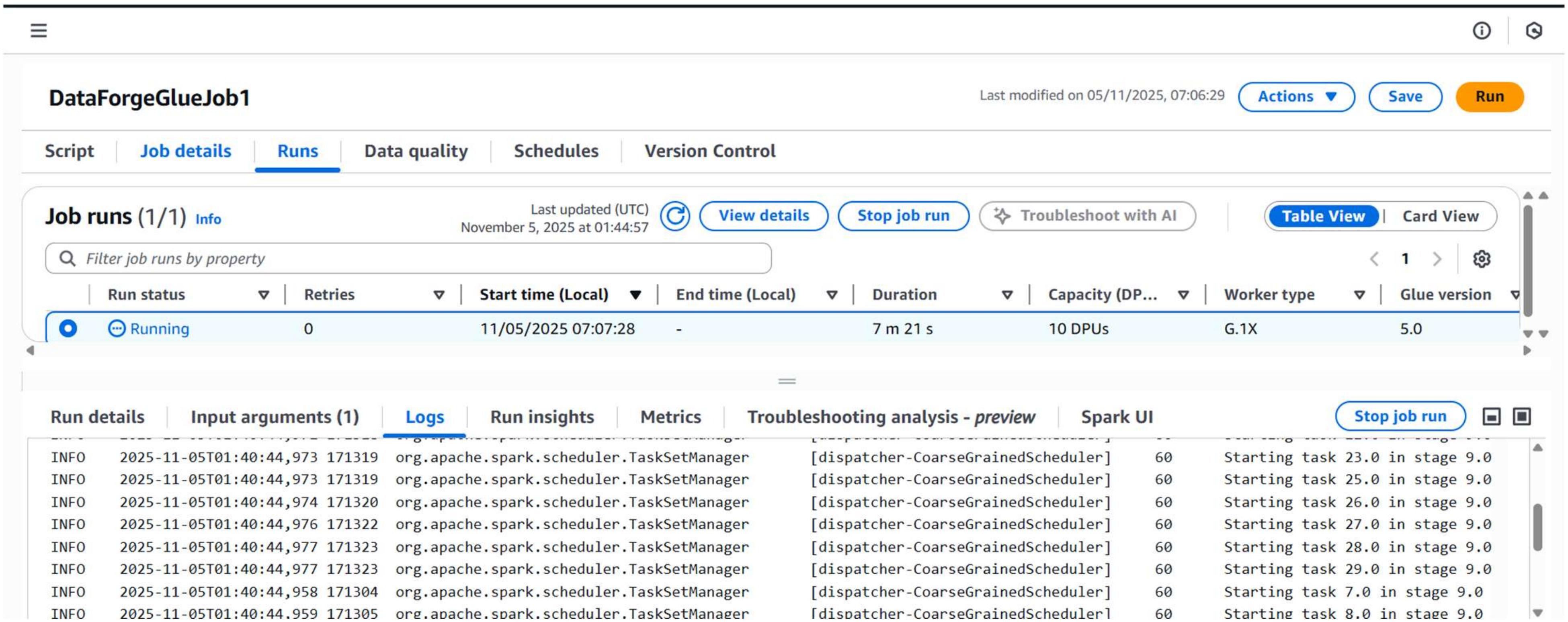

```
try:
    response = kinesis_client.create_stream(
        StreamName=stream_name,
        ShardCount=shard_count
    )
    print(f"Creating Kinesis stream '{stream_name}' with {shard_count} shard(s)...")
except ClientError as e:
        print(f"Error creating stream: {e}")
        exit(1)
```

AWS Glue Job

```
try:
    response = glue.create_job(
        Name=job_name,
        Role=role_name,
        ExecutionProperty={
            'MaxConcurrentRuns': 1
        },
        Command={
            'Name': 'glueetl',
            'ScriptLocation': script_s3_path,
            'PythonVersion': '3'
        },
        DefaultArguments={
            '--extra-jars': dependent_jars
        },
        GlueVersion='5.0',
        Description='Streaming Glue job for DataForge project',
        Tags={
            'Project': 'DataForge',
            'Type': 'StreamingJob'
        }
    )
```



AWS Glue Streaming ETL Job (Status)



KPI's

- Average signal by operator

#	operator	Avg_Signal
1	Eroski Movil	13.214285714285714
2	simyo	13.636363636363637
3	RACC	14.941176470588236
4	vodafone ES	20.17241379310345
5	Tuenti	11.038461538461538
6	GETESA	15.26086956521739
7	DTAC	14.263157894736842
8	Movistar	15.074074074074074
9	JAZZTEL	14.541666666666666
10	Orange	16.0
11	VIVO	14.805555555555555
12	YOIGO	13.625

- Average signal by postal code

#	postal_code	Avg_Signal
1	15673	21.0
2	15680	9.0
3	15707	21.0
4	15722	8.0
5	15736	27.0
6	15756	18.0
7	15792	22.0
8	15821	22.0
9	15955	21.0
10	16216	17.0

KPI's

- Average signal by operator and postal code

#	operator	postal_code	Avg_Signal
1	RACC	16328.	10.0
2	GETESA	16445.	20.0
3	VIVO	16482.	3.0
4	VIVO	16636.	27.0
5	RACC	16648.	28.0
6	DTAC	16809.	24.0
7	VIVO	16850.	30.0
8	Eroski Movil	16940.	10.0
9	Tuenti	17005.	8.0
10	Movistar	17241.	17.0

- Average GPS precision by provider

#	operator	Avg_precission
1	vodafone ES	76.75882352941177
2	simyo	68.58611111111112
3	RACC	89.70333333333335
4	Eroski Movil	72.59411764705882
5	Tuenti	93.69615384615386
6	JAZZTEL	72.72142857142856
7	Movistar	75.685
8	Orange	67.90416666666665
9	GETESA	65.72
10	VIVO	88.56923076923077

KPI’s

Activity Count

#	activity	Activity_Count
1	ON_BICYCLE	119
2	UNKNOWN	126
3	TILTING	101
4	IN_VEHICLE	101
5	STILL	114
6	ON_FOOT	118

Business Output

- Real-time data availability : Incoming mobile log data instantly processed and stored in S3.
- Clean & structured datasets : JSON stream transformed into timestamped, analytics-ready CSV files.
- Faster insights : Teams can query live data using Amazon Athena without manual ETL steps.
- Scalable foundation : The pipeline can easily integrate new data sources or analytics dashboards (QuickSight, Power BI).

Key Learnings

1. AWS Data Pipeline Design : Learned how to build a real-time data pipeline connecting Kinesis Data Stream → AWS Glue → S3 → Athena for end-to-end data processing.
2. Spark Structured Streaming : Gained hands-on experience with reading continuous streams and writing results.
3. Schema Handling & Data Transformation : Practiced defining JSON schemas, parsing dynamic data, and transforming raw data into structured CSV output.
4. Offline Testing & Local Simulation : Developed a local test setup using Docker + Spark to simulate streaming with JSON files before deploying to AWS.
5. AWS Cloud Integration : Explored Glue Catalog, Athena, and S3 as a seamless ecosystem for data storage, querying, and analytics.

Future Scope

Real-Time Analytics Dashboard	Integrate with Amazon QuickSight or Grafana for live visualization of streaming data.
Data Quality & Validation Layer	Add AWS Lambda or Glue triggers to validate and cleanse data before storage.
Scalable Multi-Stream Integration	Extend the pipeline to handle multiple Kinesis streams for different data sources or applications.

Thank you